

.NET Serialization: Detecting and defending vulnerable endpoints

.NET Serialization: Detecting and defending vulnerable endpoints - @speakerdeck, Alvaro, Speaker Deck.

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Content

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PDF to Markdown

.NET SERIALIZATION

Alvaro Muñoz pwntester

> **whoami**

§ **Alvaro Muñoz a.k.a. @pwntester**

- Principal security researcher with Micro Focus Fortify
- Presented my research at different conferences such as:
- BlackHat, Defcon, RSA, OWASP AppSecEU, AppSecUSA, JavaOne, etc.
- Responsibly reported critical vulnerabilities to companies/frameworks such as:
- Microsoft, Oracle, Workday, Salesforce, HPE, Pivotal, Apache, Atlassian, Lightbend, etc.

Some serialization experience

<http://blog.diniscruz.com/2013/08/usinghttp://www.pwntester.com/blog/2013/12/23/rce-xmldecoder-via--xstreamto-execute-object-server-deserialization38/-side.html>
<http://www.pwntester.com/blog/2013/12/16/cvehttps://gist.github.com/pwntester/ab70e88821b4a6633c06-2011-2894-deserialization-spring-rce/>
3 <https://github.com/pwntester/SerialKillerBypassGadgetCollection>

XMLDecoderXStreamSpring RCE

Gafget

Apache CommonsCollections
RCE Gadget

Look

- Ahead Bypass Multiple RCE GadgetsJRE 8u20 RCE GadgetJSON Deserialization

2013 2014 2015 2016 2017 2018

...

2012

Tic

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Tac Tic

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(^101010110010101101010101011011) 1000 10101010110101000
101010110010101001

Agenda

- Serialization 101
- .NET serializers
- Native
- 3rd Party
- Detecting vulnerable endpoints
- Fixing vulnerable endpoints

Inside

- -

- Serialization

Marshalling Pickles

Marshalling Pickles

Marshalling Pickles

Marshalling Pickles

Pickle Rick

Marshalling Pickles

Pickle Rick

Marshalling Pickles Type Discriminator Pickle Rick

Marshalling Pickles

Pickle Rick

Marshalling Pickles

Pickle Rick

Marshalling Pickles Morty Pickle Rick

Marshalling Pickles

Pickle Rick

Morty

Methods Invoked to Fully Reconstruct Objects

§ Deserialization callbacks:

- Java:
- readObject/readResolve
- .NET:
- Deserialization constructor overload
- * (SerializationInfo, StreamingContext)*
- IDeserializationCallback.OnDeserialization(Object)
- [OnDeserializing]/[OnDeserialized] annotated methods

§ Setters

Gadgets

§ Attacker controls:

§ Gadget:

- Type which contains one or more methods invoked during the deserialization process that under controlled circumstances may do bad

things

Type Type Property Values

System.Windows.Data.ObjectDataProvider

```
set_MethodName()
```

```
BeginQuery()
```

```
QueryWorker()
```

```
InvokeMethodOnInstance()
```

```
Refresh()
```

```
set_ObjectTypeset_ObjectInstance() ()
```

Gadgets

ysoserial.net

.NET Formatters

Introduction

§Attacks on .NET formatters are not new §James Forshaw already introduced them at BlackHat 2012 for

- BinaryFormatter (Binary)
- NetDataContractSerializer (XML)

§Lack of Remote Code Execution gadgets until 2017

Vulnerable in default configuration

§ BinaryFormatter(Binary)

- BinaryMessageFormatter(Binary) [MSMQ]
- ObjectStateFormatter(Binary) [ViewState]
- LosFormatter(Binary)

§ NetDataContractSerializer(XML) § SoapFormatter(XML) § FastJSON (JSON) § Sweet.Jayson(JSON)

BinaryFormatter

Eg: AppHarbor

Eg: AppHarbor

Super-Cookie AntiPattern

(^31) <https://blog.appharbor.com/2012/04/04/cookietempdataprotvider-for-asp-net> (now deleted)

Actually that advice is everywhere :(

Silently removed from ASP.NET MVC

Demo

Azure Active Directory Application Proxy

Vulnerable if developers mess it up (1/2)

§Attacker can control Expected Type:

- DataContractSerializer (XML)
- DataContractJsonSerializer (JSON)
- XmlSerializer (XML)
- XmlMessageSerializer (XML) [MSMQ]

XmlSerializer

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DotNetNukeCMS (CVE- 2017 - 9822)

Do not **let** users control Expected Type

Vulnerable if developers mess it up (2/2)

§Insecure Configuration:

- JavaScriptSerializer (JSON)
- JSON.NET (JSON)
- FSPickler (JSON)

JavaScriptSerializer

Do not use Type Resolver

JSON.NET

41 Do not use TypeNameHandling != None

Detecting Vulnerable

Endpoints

Passive

§Magic numbers:

§Burp plugin

- pwntester/dotnet-deserialization-scanner
- False Positives
- Some Images may contain similar bytes
- May appear in signed ViewState

AAEAAAD/////...

Active

§Send payload and watch execute (DAST)

- Use ysoserial.net to generate:
- DoS gadget (sleep)
- URL gadget (DNS Lookup)

§Instrument deserialize methods (IAST)

- Monitor running application

Static

§Single dataflow+controlflow

- Track data to be deserialized
- eg: BinaryFormatter

§Dual dataflow+controlflow

- Track data to be deserialized and expected type

- eg: XmlSerializer

Fixing vulnerable endpoints

1 - Stop using it

1 - Stop using it

§Do you really need it?

- eg: Nancy (CVE- 2017 - 9785)
- NCSRF cookie (CSRF token)

§Do you really need Type discriminators in JSON/XML?

- eg: Breeze (CVE- 2017 - 9424)
- Type information not needed since it works with JS clients

JSON.NET

Use TypeNameHandling== None

2 - Sign and verify it

§ Use **HMAC** , never MD5(secret + data) | SHA1(secret + data) § Examples:

- AppHarbor
- Azure Active Directory

§ ASP.NET MVC Futures -> ASP.NET MVC

- Uses the DataProtectionAPI which offers both Integrity and Confidentiality

§ ASP.NET ViewState

Signed Cookie

DataProtector.Protect(bytes) == Sign it (and optionally encrypt it)

ViewState

§ViewState contains the page state serialized using ObjectStateFormatter. §Since 4.5.2 ASP.NET ignores `EnableViewStateMac` and will always sign and encrypt the ViewState

- Patch was applied retroactively back to 1.1

§Still found hundreds (**200+**) of servers using old versions without signing/encryption!

ViewState

§In 4.5 Microsoft added Purpose to derive unique keys for each request

Encryption KeyValidation Key KDF Encryption KeyValidation Key

(per-request) Purposes Strings

MachineKey (per-request) keys

ViewState

\$PrimaryPurpose and some specific purposes are easily predictable,
but what about ViewStateUserKey ...

URL: /Account/Register

ViewState

Careful with leaking the keys

\$Leak web.config through XXE vulnerabilities

- eg: AfterLogic WebMail Pro ASP.NET 6.2.6 - Administrator Account

Disclosure via XXE \$Leak web.config through Padding Oracle

- (MS10-070) (CVE- 2010 - 3332)

\$Vulnerability in .NET Framework Could Allow Information Disclosure

- (MS15-041) (CVE- 2015 - 1648)

Yellow Screen of Death

(^57) <https://www.troyhunt.com/owasp-top-10-for-net-developers-part-6/>

Don't make it public

Careful with One-Click Installers

Careful with leaking the key

[https://msdn.microsoft.com/en-us/library/ms178199\(v=vs.85\).aspx](https://msdn.microsoft.com/en-us/library/ms178199(v=vs.85).aspx)

You can help prevent modification to your application configuration by
encrypting sections of configuration files.

For more information, see “ Encrypting Configuration Information **Using Protected** Configuration” ([https://msdn.microsoft.com/en-us/library/53tyfkaw\(v=vs.85\).aspx](https://msdn.microsoft.com/en-us/library/53tyfkaw(v=vs.85).aspx))

3 - Bind it

\$Constrain allowed types \$Serialization binders

- Allows users to control class loading and mandate what class to

load.

\$Also Known As “look-ahead deserialization” in Java

Strict White List

(^62) **Credit: Jonathan Birch -Microsoft Corporation**

Strict White List

Never use BlackLists or Broad WhiteLists

Bypass Gadgets System.Data.DataSet

Also ...

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- Don't use *IsAssignableFrom*
- Attackers can find a generic Object type in the Object graph to place the payload.

- Don't *return null* for unexpected types
- Some serializers fall back to a default binder, allowing exploits.
- Don't use reflection to look up types:

Assembly.Load(assemblyName).GetType(typeName);

- Reflection is slow, and a malicious user can DoS your application by forcing it to spend memory and time loading irrelevant assemblies.

Credit: Jonathan Birch -Microsoft Corporation

4 - Replace It

§Structured Data Approaches:

- You define how you want your data to be structured once, then you can use special generated source code to easily write and read your structured data to and from a variety of data streams and using a variety of languages.

- Eg: Google Protocol Buffers

§Untyped JSON/XML

- Eg: Json.NET with TypeNameHandling.None

Mahalo!

alvaro.munoz@microfocus.com
@pwntester

Slide images preserved from the transcript links

.NET SERIALIZATION

Alvaro Muñoz

 pwntester



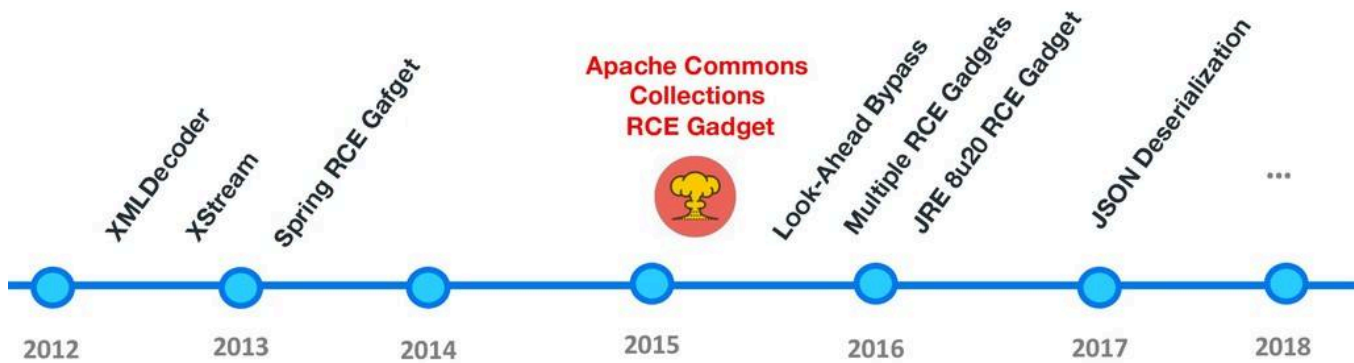
> whoami

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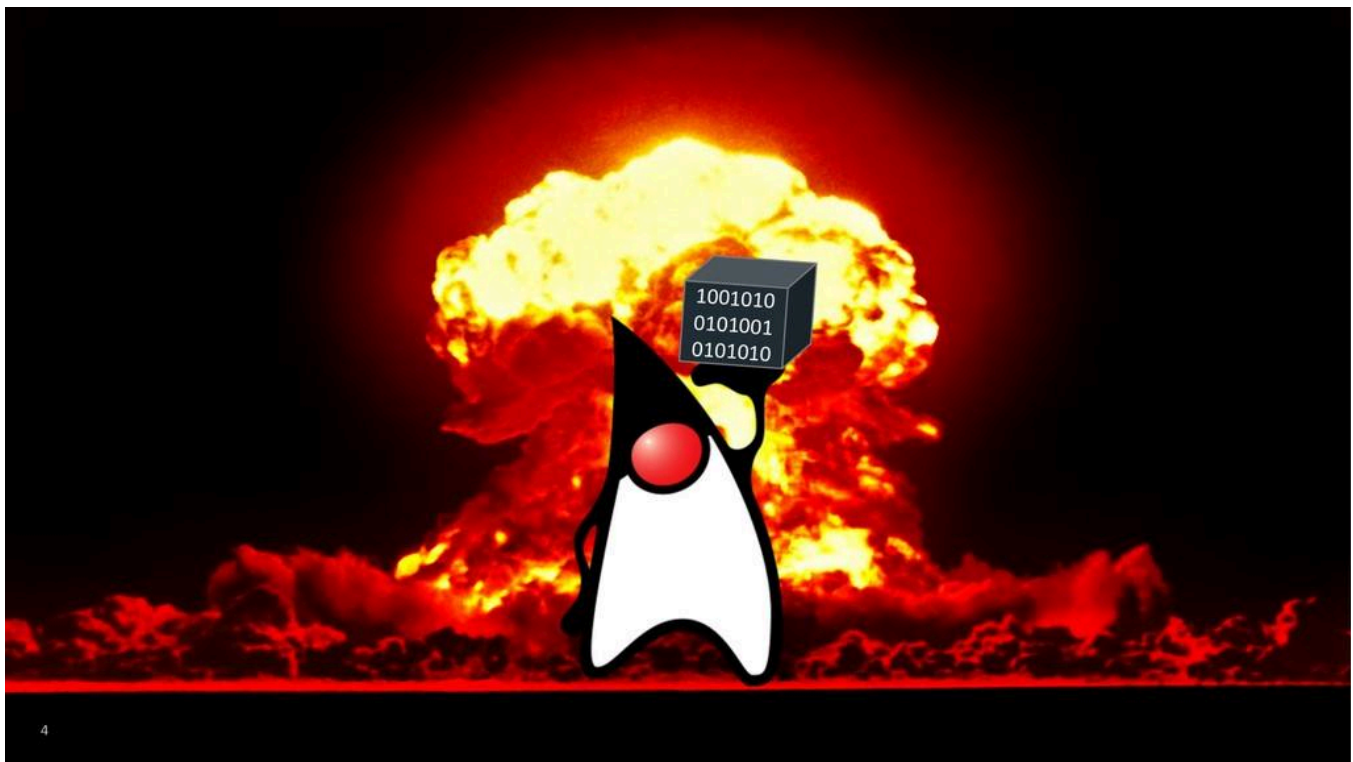
Some serialization experience



<http://blog.diniscruz.com/2013/08/using-xmldecoder-to-execute-server-side.html>
<http://www.pwntester.com/blog/2013/12/23/rce-via-xstream-object-deserialization38/>
<http://www.pwntester.com/blog/2013/12/16/cve-2011-2894-deserialization-spring-rce/>
<https://gist.github.com/pwntester/ab70e88821b4a6633c06>
<https://github.com/pwntester/SerialKillerBypassGadgetCollection>

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A8
:2017

Insecure Deserialization

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 <pre>graph LR; ThreatAgents[Threat Agents] -.-> AttackVectors[Attack Vectors]; AttackVectors -.-> SecurityWeakness[Security Weakness]; SecurityWeakness -.-> Impacts[(Impacts)];</pre>					
App. Specific	Exploitability: 1	Prevalence: 2	Detectability: 2	Technical: 3	Business ?
Exploitation of deserialization is somewhat difficult, as off the shelf exploits rarely work without changes or tweaks to the underlying exploit code.		This issue is included in the Top 10 based on an industry survey and not on quantifiable data. Some tools can discover deserialization flaws, but human assistance is frequently needed to validate the problem. It is expected that prevalence data for deserialization flaws will increase as tooling is developed to help identify and address it.		The impact of deserialization flaws cannot be understated. These flaws can lead to remote code execution attacks, one of the most serious attacks possible. The business impact depends on the protection needs of the application and data.	

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Agenda

1. Serialization 101
2. .NET serializers
 1. Native
 2. 3rd Party
3. Detecting vulnerable endpoints
4. Fixing vulnerable endpoints

DEMO
Inside

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Serialization 101

Marshalling Pickles

Marshalling Pickles



Marshalling Pickles



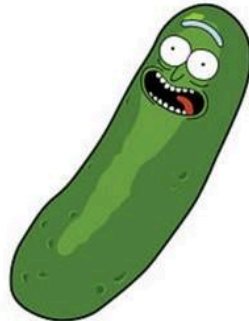
11

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Marshalling Pickles



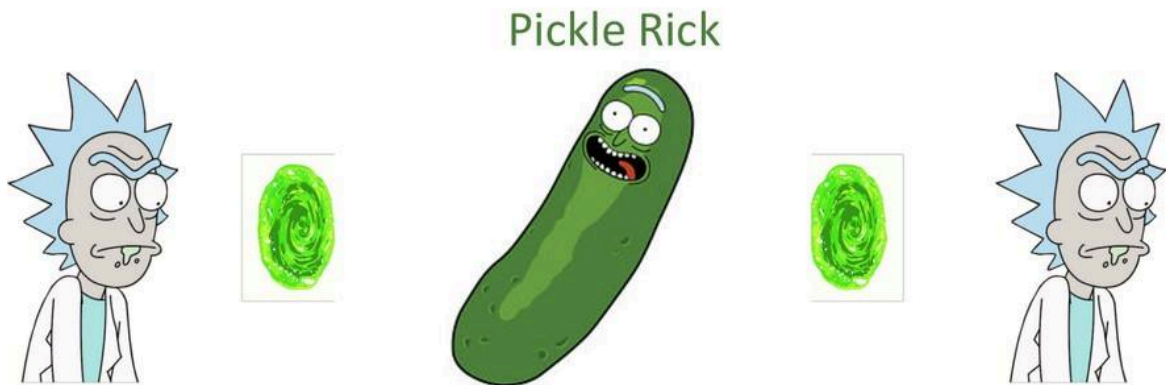
Pickle Rick



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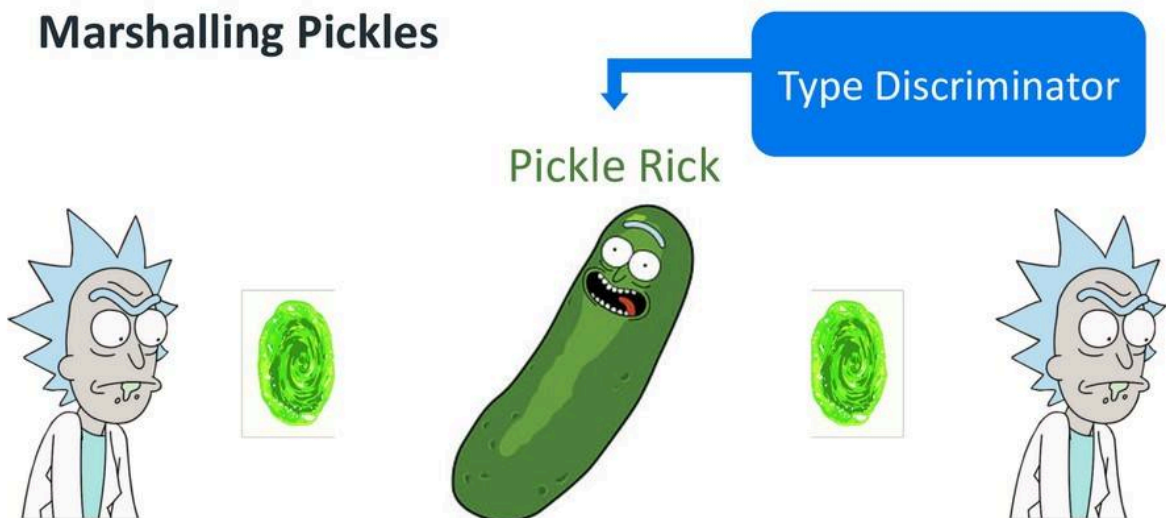
Marshalling Pickles



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Marshalling Pickles



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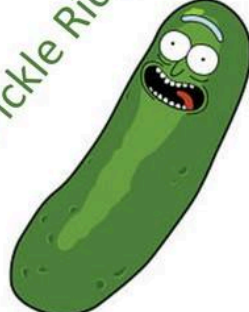
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Marshalling Pickles

```
0000000: aced 0005 7372 001d 636f 6d2e 7175 616c ....sr. com.qual
0000010: 636f 6d6d 2e69 7372 6d2e 6170 7073 6563 comm.isrm.appsec
0000020: 2e55 7365 7200 0000 0000 0000 0102 0002 .User.....
0000030: 5a00 0b75 7365 7249 7341 646d 696e 4c00 z..userIsAdmin..
0000040: 046e 616d 6574 0012 4c6a 6176 612f 6c61 .namet..Ljava/la
0000050: 6e67 2f53 7472 696e 673b 7870 0074 0004 ng/String;xp.t..
0000060: 6761 6265 gabe
```



Pickle Rick



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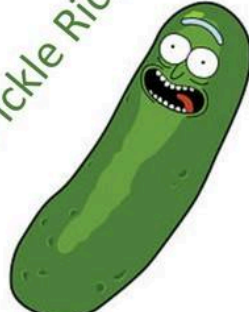
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Marshalling Pickles

```
{
  "type": "SampleApp.Types.Person, SampleApp",
  "profession": {
    "type": "SampleApp.Types.Programming, SampleApp",
    "jobTitle": "Software Developer",
    "favoriteLanguage": "C#"
  }
}
```



Pickle Rick



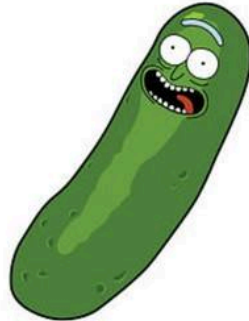
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Marshalling Pickles

Morty

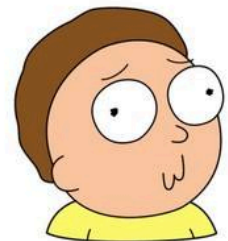
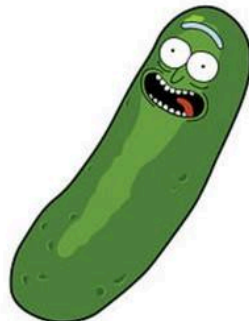
~~Pickle Rick~~



Marshalling Pickles

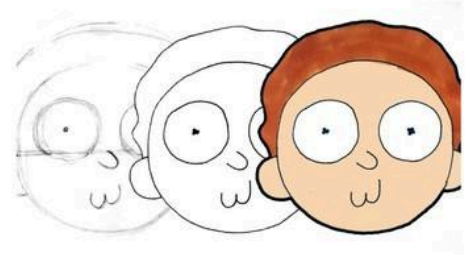
Morty

~~Pickle Rick~~



Methods Invoked to Fully Reconstruct Objects

- Deserialization callbacks:
 - Java:
 - readObject/readResolve
 - .NET:
 - Deserialization constructor overload
 - `<Type> (SerializationInfo, StreamingContext)`
 - `IDeserializationCallback.OnDeserialization(Object)`
 - `[OnDeserializing]/[OnDeserialized]` annotated methods
- Setters

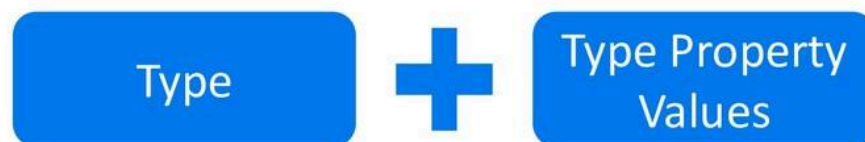


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Gadgets

- Attacker controls:

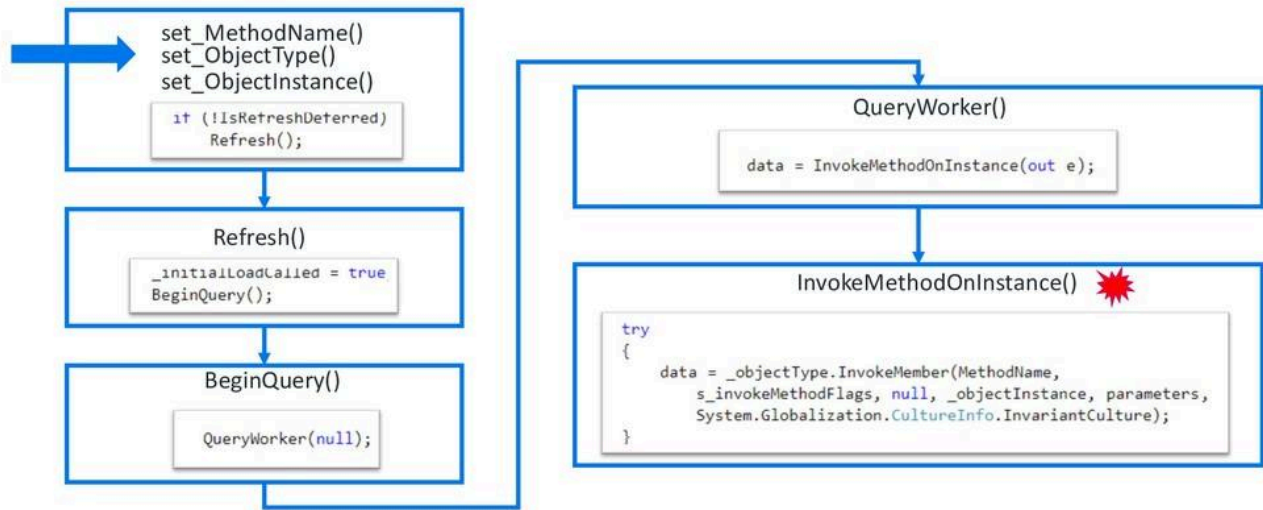


- Gadget:
 - Type which contains one or more methods invoked during the deserialization process that under controlled circumstances may do bad things

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System.Windows.Data.ObjectDataProvider



Gadgets

```
{
  "$type": "System.Windows.Data.ObjectDataProvider, PresentationFramework",
  "ObjectInstance": {
    "$type": "System.Diagnostics.Process, System"
  },
  "MethodParameters": {
    "$type": "System.Collections.ArrayList, mscorlib",
    "$values": ["calc"]
  },
  "MethodName": "Start"
}
```

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.NET Formatters

Introduction

- Attacks on .NET formatters are not new
- James Forshaw already introduced them at BlackHat 2012 for
 - BinaryFormatter (Binary)
 - NetDataContractSerializer (XML)
- Lack of Remote Code Execution gadgets until 2017

Vulnerable in default configuration

- BinaryFormatter (Binary)
 - BinaryMessageFormatter (Binary) [MSMQ]
 - ObjectStateFormatter (Binary) [ViewState]
 - LosFormatter (Binary)
- NetDataContractSerializer (XML)
- SoapFormatter (XML)
- FastJSON (JSON)
- Sweet.Jayson (JSON)



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BinaryFormatter

```
var cookie = new HttpCookie(CookieName);
var formatter = new BinaryFormatter();
using (var stream = new MemoryStream(data))
{
    formatter.Serialize(stream, values);
    var bytes = stream.ToArray();
    cookie.Value = Convert.ToBase64String(bytes);
}
controllerContext.HttpContext.Response.Cookies.Add(cookie);
```

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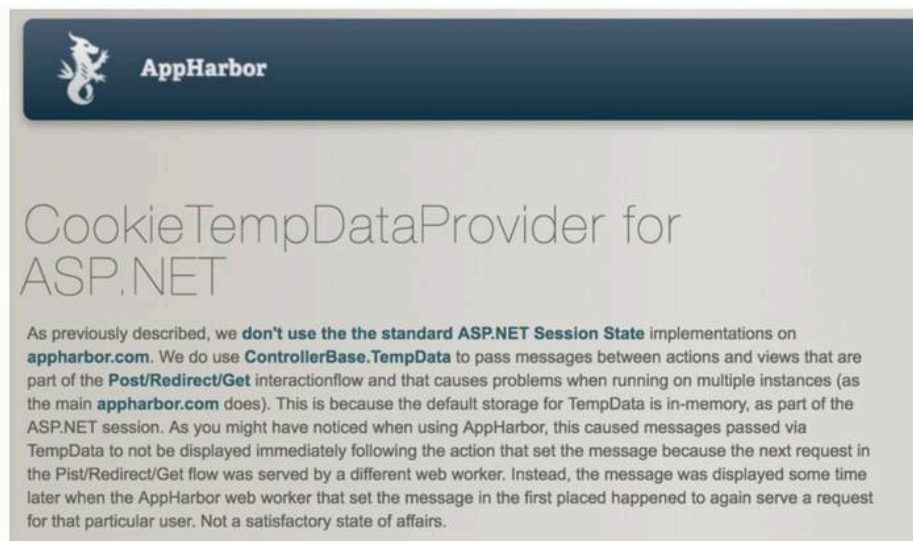
Eg: AppHarbor

The screenshot shows the AppHarbor web application interface. On the left, there's a sidebar with navigation links: Add-ons, Certificates, Collaborators, Configuration variables, Errors, Hostnames, Labs, Logging, Service hooks, Settings, and Subscription. The main content area is titled 'Application test2' and includes sections for 'Getting started' and 'Installed add-ons'. On the right, a browser's developer console is open, displaying the 'Network' tab. It shows a request to 'https://appharbor.com/applications' with a status code of 200. The response headers are visible, including 'cache-control: private, s-maxage=0', 'content-length: 139', 'content-type: text/html; charset=utf-8', 'date: Thu, 30 Mar 2017 17:04:51 GMT', 'location: /applications/test2-93', and 'server: nginx'. The response body is partially visible, showing a 'set-cookie' header.

Eg: AppHarbor

This screenshot is similar to the one above, but with a magnifying glass effect applied to the browser's developer console. The magnifying glass is positioned over the 'Network' tab, highlighting the response headers for the request to 'https://appharbor.com/applications'. The highlighted text includes: 'cache-control: private, s-maxage=0', 'content-length: 139', 'content-type: text/html; charset=utf-8', 'date: Thu, 30 Mar 2017 17:04:51 GMT', 'location: /applications/test2-93', and 'server: nginx'. The 'set-cookie' header is also visible, showing 'TempData=AAAAAAD/////AQAAAAAAAAA'.

Super-Cookie AntiPattern

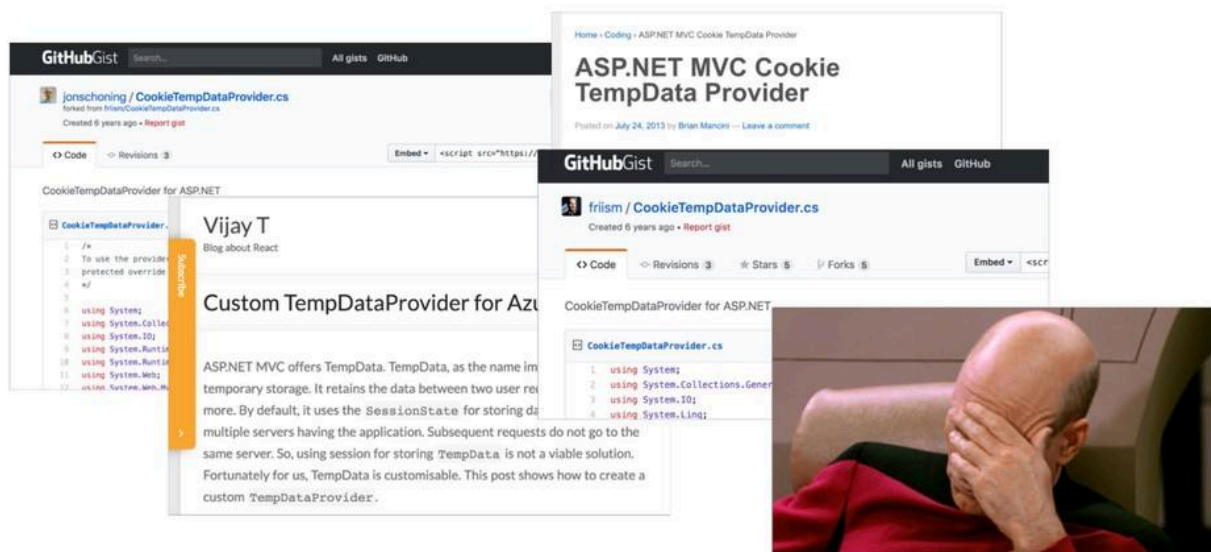


31

<https://blog.appharbor.com/2012/04/04/cookietempdataprovder-for-asp-net> (now deleted)



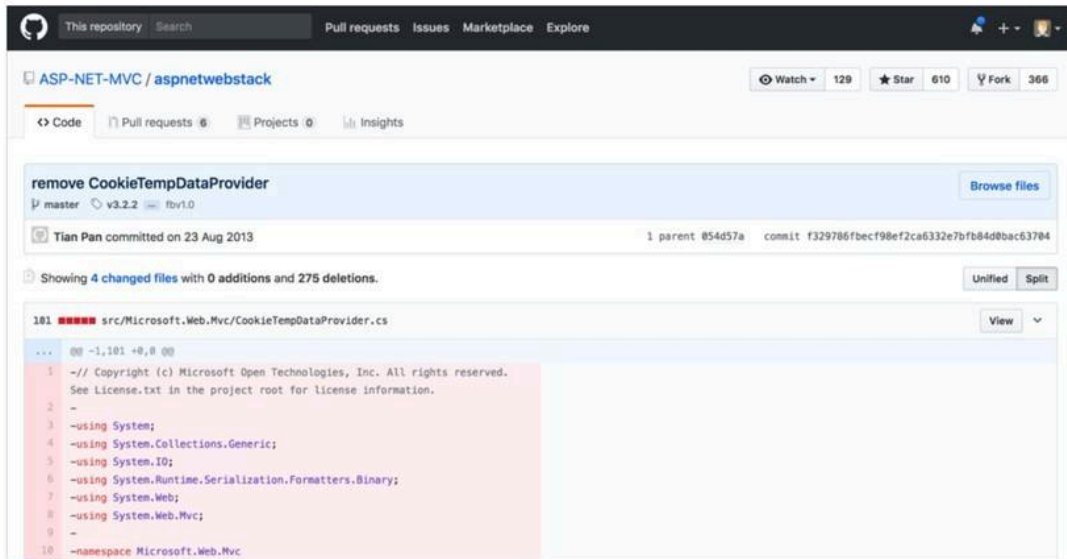
Actually that advice is everywhere :(



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Silently removed from ASP.NET MVC



ASP.NET-MVC / aspnetwebstack

remove CookieTempDataProvider

master v3.2.2 f329786fbecf98ef2ca6332e7bfb84d8bac63784

Tian Pan committed on 23 Aug 2013

Showing 4 changed files with 0 additions and 275 deletions.

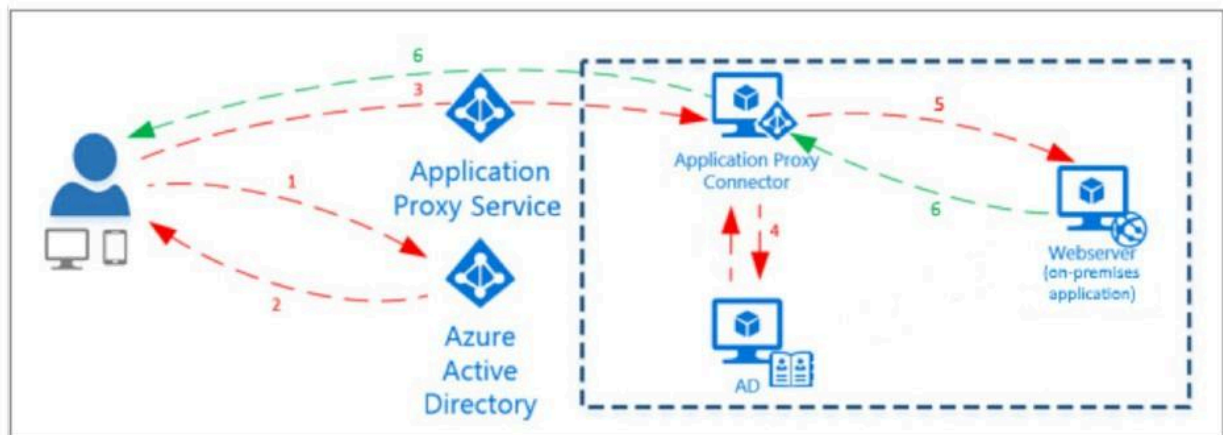
```
101 src/Microsoft.Web.Mvc/CookieTempDataProvider.cs
... -1,101 +0,0 ...
1 // Copyright (c) Microsoft Open Technologies, Inc. All rights reserved.
2 // See License.txt in the project root for license information.
3
4 -using System;
5 -using System.Collections.Generic;
6 -using System.IO;
7 -using System.Runtime.Serialization.Formatters.Binary;
8 -using System.Web;
9 -using System.Web.Mvc;
10 -namespace Microsoft.Web.Mvc
```

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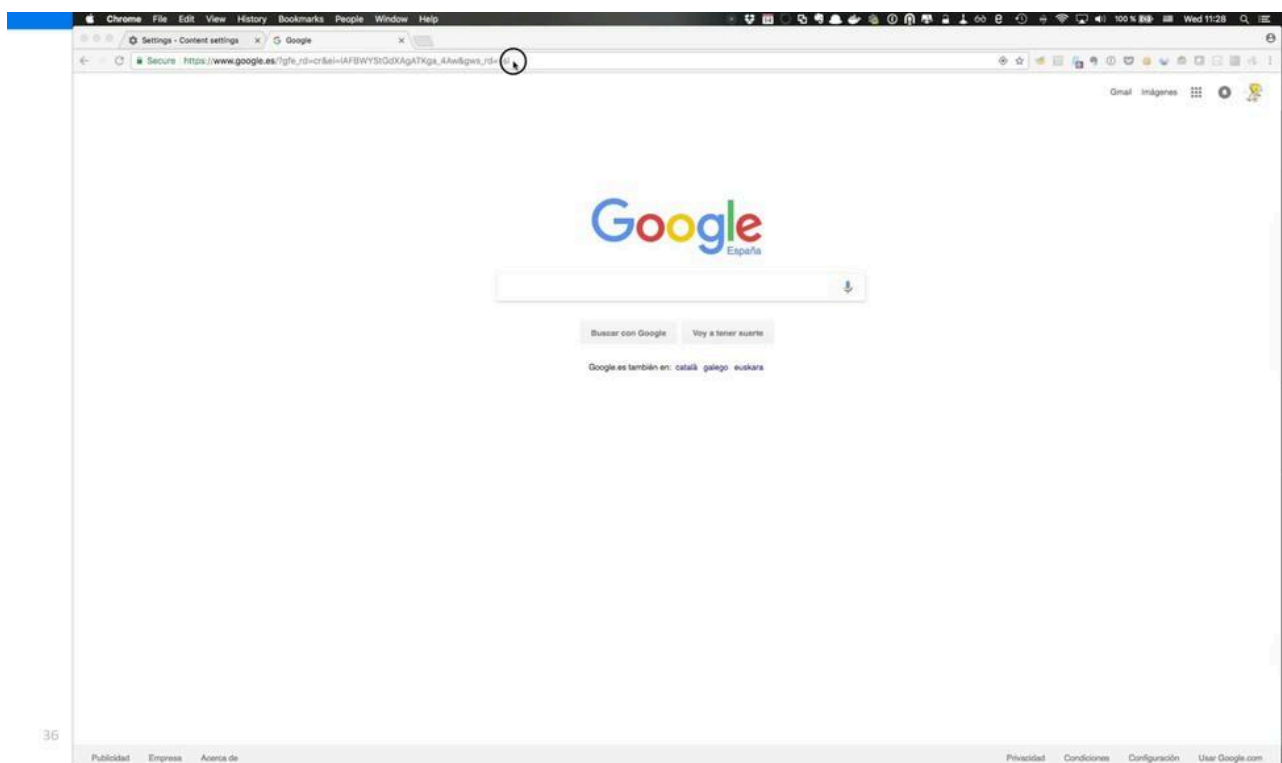
Demo



Azure Active Directory Application Proxy



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Vulnerable if developers mess it up (1/2)

- Attacker can control Expected Type:

- DataContractSerializer (XML)
- DataContractJsonSerializer (JSON)
- XmlSerializer (XML)
- XmlMessageSerializer (XML) [MSMQ]



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XmlSerializer

DotNetNuke CMS (CVE-2017-9822)

```
// xmlItem comes from an unsigned cookie
string key = xmlItem.GetAttribute("key");
string typeName = xmlItem.GetAttribute("type");

//Create the XmlSerializer
var xser = new XmlSerializer(Type.GetType(typeName));

//A reader is needed to read the XML document.
var reader = new XmlTextReader(new StringReader(xmlItem.InnerXml));

//Use the Deserialize method to restore the object's state, and store it
//in the Hashtable
hashTable.Add(key, xser.Deserialize(reader));
```

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Do not let users control Expected Type

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Vulnerable if developers mess it up (2/2)

- Insecure Configuration:
 - JavaScriptSerializer (JSON)
 - JSON.NET (JSON)
 - FSPickler (JSON)



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JavaScriptSerializer

```
JavaScriptSerializer sr = new JavaScriptSerializer(new SimpleTypeResolver());  
string reqdInfo = apiService.authenticateRequest();  
reqdDetails det = (reqdDetails)(sr.Deserialize<reqdDetails>(reqdInfo));
```

Do not use Type Resolver

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JSON.NET

```
var jsonSerializerSettings = new JsonSerializerSettings() {  
    NullValueHandling = NullValueHandling.Include,  
    PreserveReferencesHandling = PreserveReferencesHandling.Objects,  
    ReferenceLoopHandling = ReferenceLoopHandling.Ignore,  
    TypeNameHandling = TypeNameHandling.Objects  
    TypeNameAssemblyFormat = FormatterAssemblyStyle.Simple,  
};
```

Do not use `TypeNameHandling != None`

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**Detecting Vulnerable
Endpoints**



Passive

- Magic numbers: **AAEAAAD/////...**
- Burp plugin
 -  pwntester/dotnet-deserialization-scanner
 - False Positives
 - Some Images may contain similar bytes
 - May appear in signed ViewState

Active

- Send payload and watch execute (DAST)
 - Use ysoserial.net to generate:
 - DoS gadget (sleep)
 - URL gadget (DNS Lookup)
- Instrument deserialize methods (IAST)
 - Monitor running application

Static

- Single dataflow+controlflow
 - Track data to be deserialized
 - eg: BinaryFormatter
- Dual dataflow+controlflow
 - Track data to be deserialized and expected type
 - eg: XmlSerializer

45

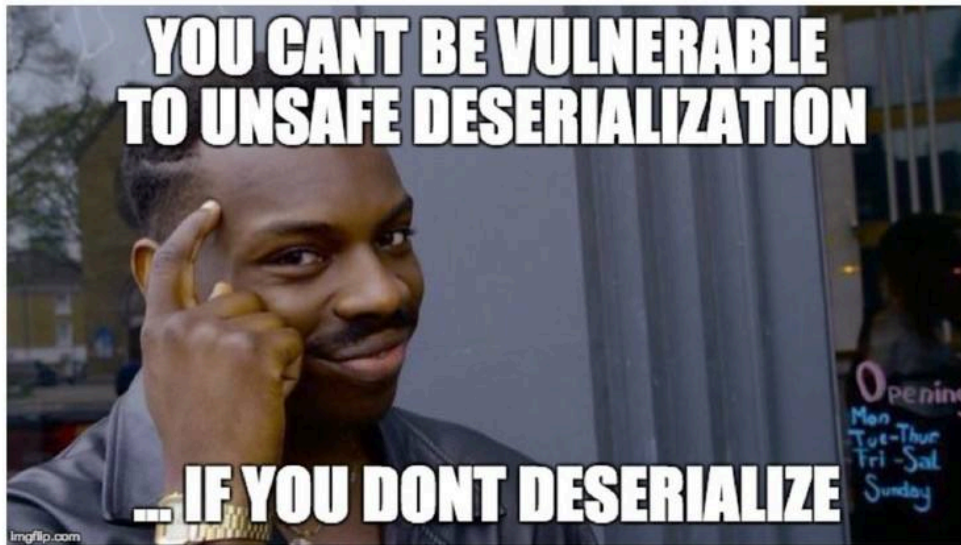
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Fixing vulnerable endpoints

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1 - Stop using it



1 - Stop using it

- Do you really need it?
 - eg: Nancy (CVE-2017-9785)
 - NCSRF cookie (CSRF token)
- Do you really need Type discriminators in JSON/XML?
 - eg: Breeze (CVE-2017-9424)
 - Type information not needed since it works with JS clients

JSON.NET

```
var jsonSerializerSettings = new JsonSerializerSettings() {  
    // DO NOT CHANGE THIS SETTING, EVER!  
    TypeNameHandling = TypeNameHandling.None  
};
```

Use `TypeNameHandling == None`

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2 - Sign and verify it

- Use **HMAC**, never **MD5(secret + data) | SHA1(secret + data)**
- Examples:
 - AppHarbor
 - Azure Active Directory
- ASP.NET MVC Futures -> ASP.NET MVC
 - Uses the DataProtection API which offers both Integrity and Confidentiality
- ASP.NET ViewState

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Signed Cookie

```
var bytes = tempDataSerializer.Serialize(values);  
bytes = _dataProtector.Protect(bytes);  
var encodedValue = Base64UrlTextEncoder.Encode(bytes);
```

`DataProtector.Protect(bytes)` == Sign it (and optionally encrypt it)

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ViewState

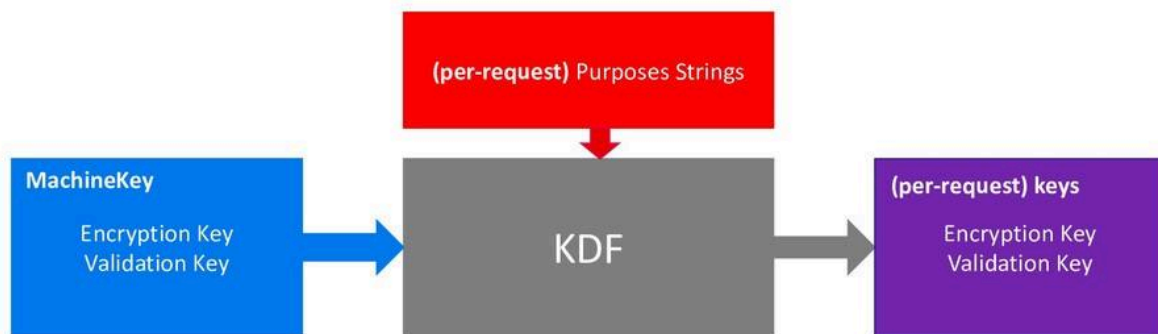
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- Since 4.5.2 ASP.NET ignores `EnableViewStateMac` and will always sign and encrypt the ViewState
 - Patch was applied retroactively back to 1.1
- Still found hundreds (**200+**) of servers using old versions without signing/encryption!

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ViewState

- In 4.5 Microsoft added Purpose to derive unique keys for each request



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ViewState

URL: /Account/Register

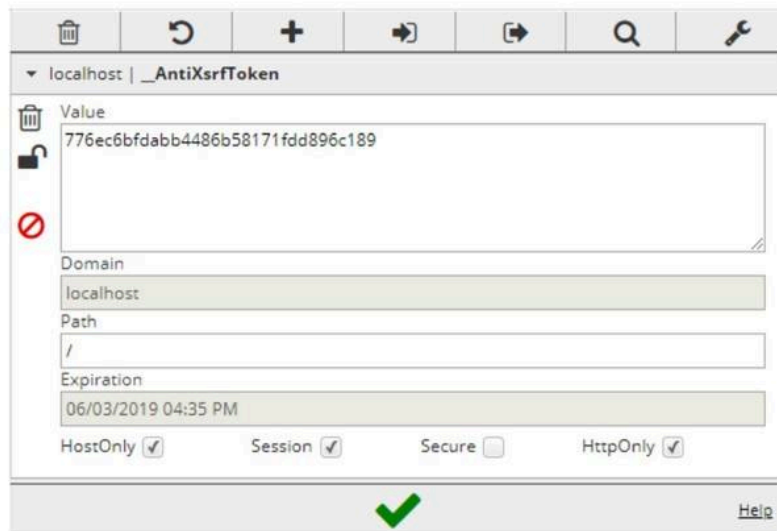
PrimaryPurpose	"WebForms.HiddenFieldPageStatePersister.ClientState"
SpecificPurposes	string[0x00000003]
[0]	"TemplateSourceDirectory: /ACCOUNT"
[1]	"Type: ACCOUNT_REGISTER_ASPX"
[2]	"ViewStateUserKey: 776ec6bfdabb4486b58171 added896c189"

- PrimaryPurpose and some specific purposes are easily predictable, but what about *ViewStateUserKey* ...

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ViewState



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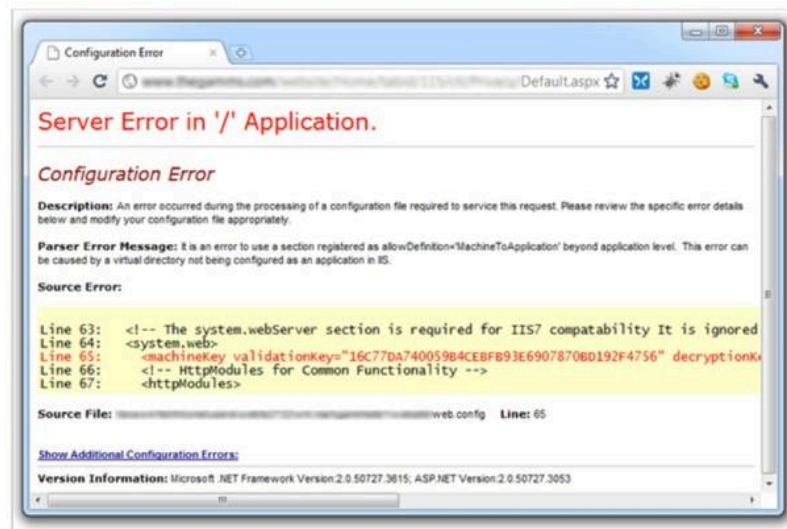
Careful with leaking the keys

- Leak web.config through XXE vulnerabilities
 - eg: AfterLogic WebMail Pro ASP.NET 6.2.6 - Administrator Account Disclosure via XXE
- Leak web.config through Padding Oracle
 - (MS10-070) (CVE-2010-3332)
- Vulnerability in .NET Framework Could Allow Information Disclosure
 - (MS15-041) (CVE-2015-1648)

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Yellow Screen of Death



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<https://www.troyhunt.com/owasp-top-10-for-net-developers-part-6/>



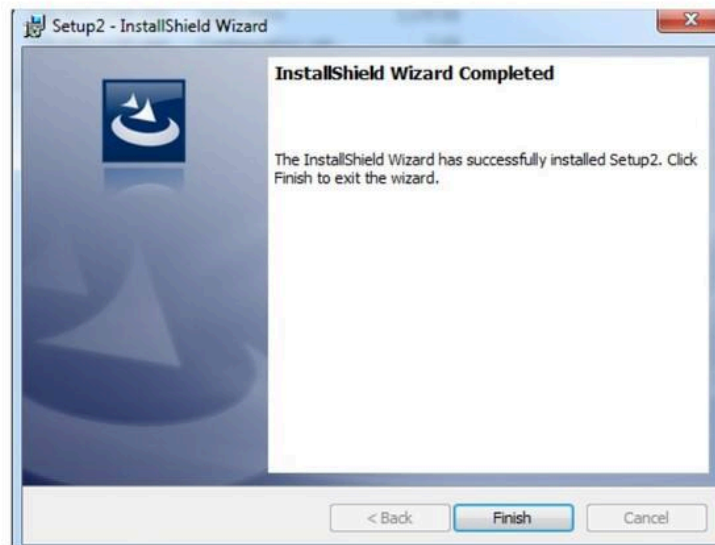
Don't make it public

```
23 lines (22 sloc) | 945 Bytes
Raw Blame History
1 <?xml version="1.0"?>
2 <configuration>
3   <configSections>
4     <section name="magicAjax" type="MagicAjax.Configuration.MagicAjaxSectionHandler, MagicAjax"/>
5   </configSections>
6   <appSettings>
7     <add key="WebDAL" value="WC.DAL"/>
8     <add key="OLEDBCONNECTIONSTRING" value="provider=microsoft.jet.oledb.4.0;data source="/>
9     <add key="dbPath" value="~/App_Data/DataBase.config"/>
10  </appSettings>
11  <system.web>
12    <machineKey validationKey="5CF1D127BC05322500320A7B55FA692BC02AFE8" decryptionKey="270450AB36318B344C926B506C9
13    <pages validateRequest="false"/>
14    <httpRuntime maxRequestLength="123960" executionTimeout="675"/>
15    <httpModules>
16      <add name="MagicAjaxModule" type="MagicAjax.MagicAjaxModule, MagicAjax"/>
17    </httpModules>
18    <compilation debug="true"/>
19    <customErrors mode="Off"/>
20    <globalization requestEncoding="utf-8" responseEncoding="utf-8"/>
21  </system.web>
22 </configuration>
```

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Careful with One-Click Installers



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Careful with leaking the key



You can help prevent modification to your application configuration by encrypting sections of configuration files.

For more information, see **"Encrypting Configuration Information Using Protected Configuration"** ([https://msdn.microsoft.com/en-us/library/53tyfkaw\(v=vs.85\).aspx](https://msdn.microsoft.com/en-us/library/53tyfkaw(v=vs.85).aspx))

[https://msdn.microsoft.com/en-us/library/ms178199\(v=vs.85\).aspx](https://msdn.microsoft.com/en-us/library/ms178199(v=vs.85).aspx)

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3 - Bind it

- Constrain allowed types
- Serialization binders
 - Allows users to control class loading and mandate what class to load.

```
C# Copy
public abstract Type BindToType (string assemblyName, string typeName);
```

- Also Known As “look-ahead deserialization” in Java

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Strict White List

```
sealed class AllowListSerializationBinder : SerializationBinder {
    List<Tuple<string, Type>> allowedTypes = new List<Tuple<string, Type>>()
    { new Tuple<string, Type>("MyType", typeof(MyType)) };

    public override Type BindToType(string assemblyName, string typeName) {
        foreach(Tuple<string, Type> typeTuple in allowedTypes) {
            if(typeName == typeTuple.Item1) {
                return typeTuple.Item2;
            }
        }
        throw new ArgumentOutOfRangeException("Disallowed type encountered");
    }
}
```

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Credit: Jonathan Birch - Microsoft Corporation



Strict White List

```
var myBinaryFormatter = new BinaryFormatter();  
myBinaryFormatter.Binder = new AllowListSerializationBinder();  
myBinaryFormatter.Deserialize(stream);
```

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Never use BlackLists or Broad WhiteLists

```
sealed class UnsafeDeserializationBinder : SerializationBinder  
{  
    public override Type BindToType(string assemblyName, string typeName)  
    {  
        Type typeToDeserialize = null;  
        if (typeName.StartsWith("Microsoft.#####"))  
        {  
            typeToDeserialize = Assembly.Load(assemblyName).GetType(typeName);  
        }  
        return typeToDeserialize;  
    }  
}
```

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Bypass Gadgets

System.Data.DataSet

```
DataSet(SerializationInfo info, StreamingContext context)
> DataSet(SerializationInfo info, StreamingContext context, bool constructSchema)
>> DeserializeDataSet(info, context, remotingFormat, schemaSerializationMode)
>>> DeserializeDataSetSchema (info, context, remotingFormat, schemaSerializationMode)

for (int i = 0, i < tableCount; i++) {
    Byte[] buffer = (Byte[]) info.GetValue(
        String.Format(CultureInfo.InvariantCulture, "DataSet.Tables_{0}", i), typeof(Byte[]))
    );
    MemoryStream memstream = new MemoryStream(buffer);
    memStream.Position = 0;
    BinaryFormatter bf = new BinaryFormatter(null, new StreamingContext(context.State, false));
    DataTable dt = (DataTable) bf.Deserialize(memStream);
    Table.Add(dt);
}
```

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Also ...

- Don't use *IsAssignableFrom*
 - Attackers can find a generic Object type in the Object graph to place the payload.
- Don't *return null* for unexpected types
 - Some serializers fall back to a default binder, allowing exploits.
- Don't use reflection to look up types:
 - Assembly.Load(assemblyName).GetType(typeName) ;**
 - Reflection is slow, and a malicious user can DoS your application by forcing it to spend memory and time loading irrelevant assemblies.

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Credit: Jonathan Birch - Microsoft Corporation



4 - Replace It

- Structured Data Approaches:

- You define how you want your data to be structured once, then you can use special generated source code to easily write and read your structured data to and from a variety of data streams and using a variety of languages.

- Eg: Google Protocol Buffers

- Untyped JSON/XML

- Eg: Json.NET with TypeNameHandling.None

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Mahalo!



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 @pwntester

